

Eenheid 6.2

Note Title

2015/02/19

Vektore in komponentvorm Stewart p 798

Notasie:

Punt in $\mathbb{R}^2$: $P(a,b)$ koördinate

Vektor in $\mathbb{R}^2$: $\langle a, b \rangle$ komponente

• Definisie :

in Vektor is 'n hoeveelheid wat lengte en rigting het.

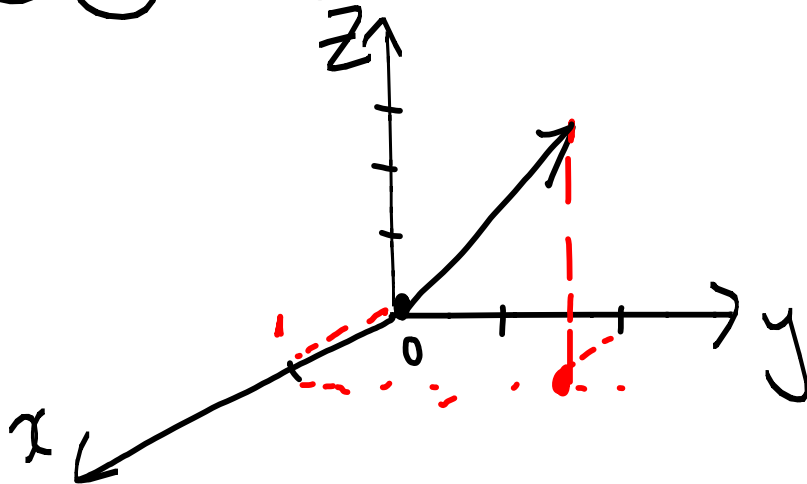
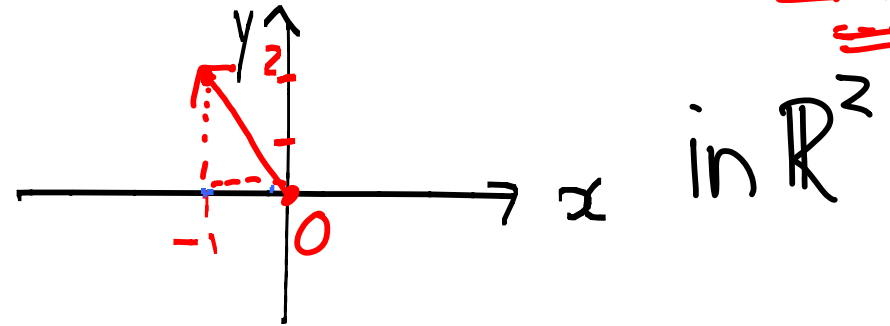
Opmerking: in Vektor het nie 'n vaste posisie nie

Notasie: in Vektor van punt A na
na punt B word geskryf as: $\overrightarrow{AB}$
= $\langle a, b, c \rangle$

Definisie: Dus die vektor van die oorsprong
 $P(0,0)$ na die punt $A(a,b)$ word geskryf as:
 $\overrightarrow{PA} = \langle a-0, b-0 \rangle = \langle a, b \rangle$

• VBD ①: $\overrightarrow{a} = \langle -1, 2 \rangle$

• VBD ②: $\overrightarrow{b} = \langle 1, 2, 3 \rangle$



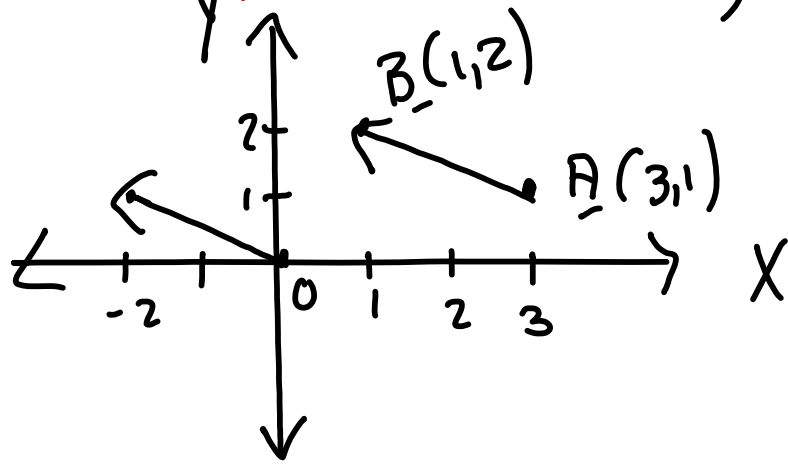
Definisie: As $A(x_1, y_1, z_1)$ en $B(x_2, y_2, z_2)$ punte is in $\mathbb{R}^3$, dan is

$$\overrightarrow{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

VBD: As $A(3,1)$ en $B(1,2)$ punte is, dan is

$$\overrightarrow{AB} = \langle 1 - 3, 2 - 1 \rangle = \langle -2, 1 \rangle$$

Posisie veld



Definisie: As $\vec{a} = \langle a_1, a_2, a_3 \rangle$ en $\vec{b} = \langle b_1, b_2, b_3 \rangle$ dan is

(vir $\mathbb{R}^3$)

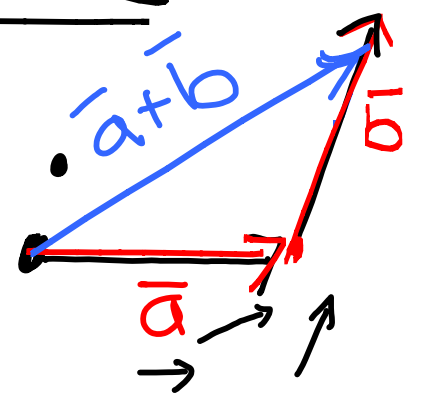
- $\vec{a} \oplus \vec{b} = \langle a_1 \oplus b_1, a_2 \oplus b_2, a_3 \oplus b_3 \rangle$
- $\vec{a} \ominus \vec{b} = \langle a_1 \ominus b_1, a_2 \ominus b_2, a_3 \ominus b_3 \rangle$

• $c\bar{a} = \langle c\bar{a}_1, c\bar{a}_2, c\bar{a}_3 \rangle \quad c \in \mathbb{R}$

① * Dieselfde geld vir $\mathbb{R}^2$

• Vektoroptelling

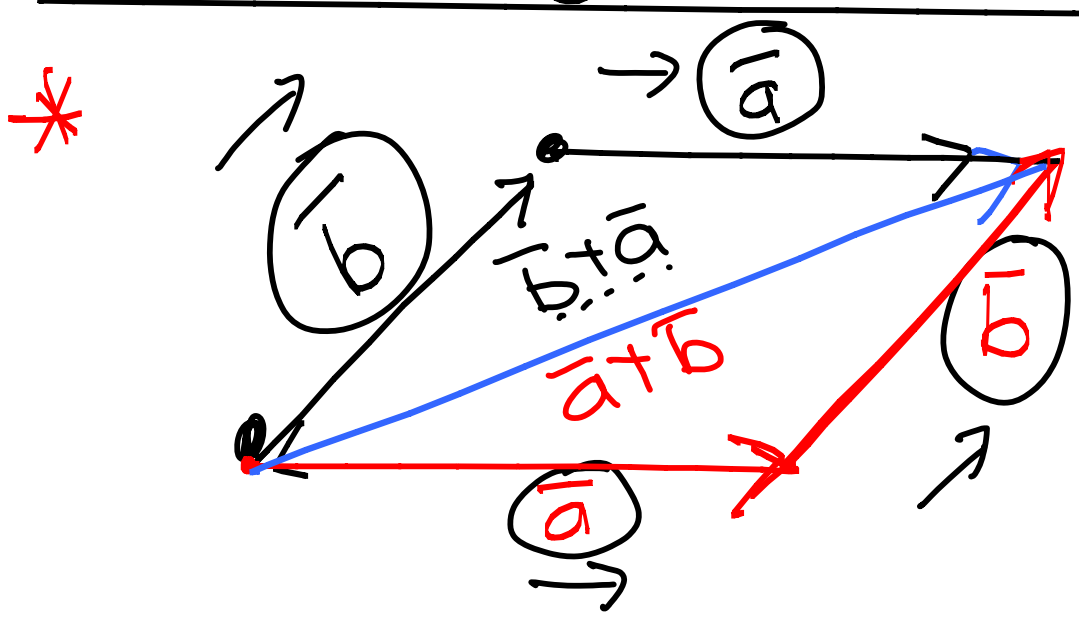
① * $\bar{a} + \bar{b}$



Stewart p789

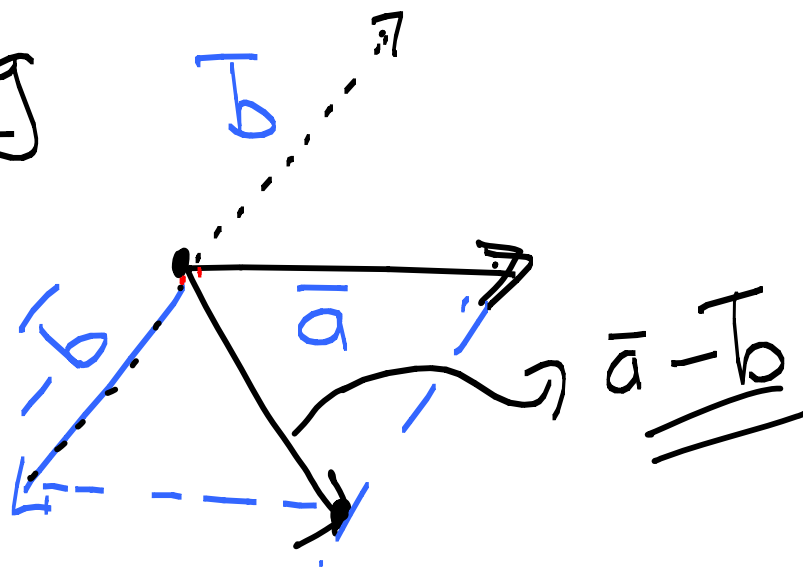
• Driehoekswet (Vektoroptelling)

• Parallellogram Wet



$$\bar{a} + \bar{b} = \bar{b} + \bar{a}$$

* Vektoraftrekking



VBD: A(2, -1, 0) en B(-3, 4, 1) en C(1, 2, -2)
punte in $\mathbb{R}^3$ is, bepaal $2\overline{AB} - 3\overline{CA}$

$$\begin{aligned}
 \cancel{B} \quad 2\overline{AB} - 3\overline{CA} &= 2 \langle -3-2, 4+1, 1-0 \rangle - 3 \langle 1-2, 2-(-1), -2-0 \rangle \\
 &= 2 \langle -5, 5, 1 \rangle - 3 \langle 1, -3, -2 \rangle \\
 &= \langle -10, 10, 2 \rangle + \langle -3, 9, -6 \rangle \\
 &= \langle -13, 19, -4 \rangle
 \end{aligned}$$

* Definitie: $\vec{a}$ en $\vec{b}$ is parallel / ewewydig as daar 'n $c \in \mathbb{R}$ bestaan sodat $\vec{a} = c\vec{b}$

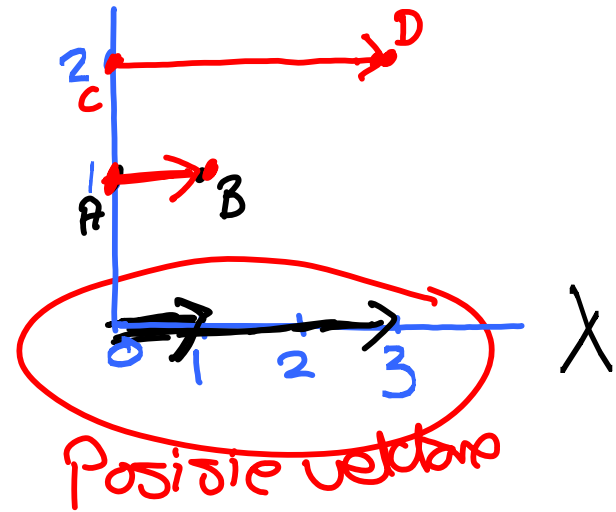
VBD: As $A(0,1)$, $B(1,1)$, $C(0,2)$ en $D(3,2)$ punte is in $\mathbb{R}^2$, is $\overline{AB} \parallel \overline{CD}$?

$$\overline{AB}_1 = \langle \underline{1}, 0 \rangle$$

$$\overline{CD}_3 = \langle \underline{3}, 0 \rangle$$

$$\text{Dan } \overline{CD} = (3) \langle \underline{1}, 0 \rangle = 3 \overline{AB}$$

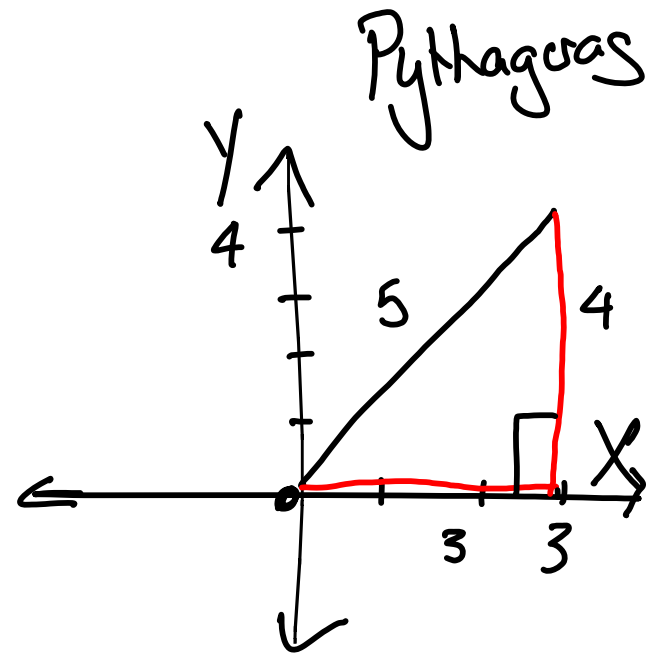
$$\text{Dus } \overline{CD} \parallel \overline{AB}$$



*Definisie: Die lengte van in vektor $\vec{a}$
gedefinieer as $|\vec{a}| = |\langle \underline{a_1}, \underline{a_2}, \underline{a_3} \rangle|$
 $= \sqrt{(a_1)^2 + (a_2)^2 + (a_3)^2}$ in $\mathbb{R}^3$

Dieselfde vir $\mathbb{R}^2$

$$\begin{aligned} \bullet |\langle \underline{3}, \underline{4} \rangle| &= \sqrt{3^2 + 4^2} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$



Definitie: in Eenheidsvektor is in
vektor met lengte 1

• $\bar{a} = \langle \underline{1, 1, 1} \rangle$ Is $\bar{a}$ in eenheidsvektor? Nee
want $|\bar{a}| = \sqrt{3} \neq 1$

• $\bar{a} = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$ Lengte van $\bar{a} = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2}$
 $= \sqrt{\frac{1}{2} + \frac{1}{2}}$

Is $\bar{a}$ in eenheidsvektor? Ja, $|\bar{a}| = 1$

Opmerking: $\bar{0}$ word die nulvektor
genoem $\bar{0} = \langle 0, 0 \rangle$ in $\mathbb{R}^2$
 $\bar{0} = \langle 0, 0, 0 \rangle$ in $\mathbb{R}^3$

- in Eenheidsvektor in dezelfde richting als $\bar{a}$ ($\bar{a} \neq 0$) is $\frac{1}{|\bar{a}|} \bar{a}$

VOORBEELD: Bepaal in vektor $\vec{b}$ met lengte 2 in die teencorgestelde richting van $\bar{a} = \langle -3, 4 \rangle$

- $|\bar{a}| = \sqrt{25} = 5$

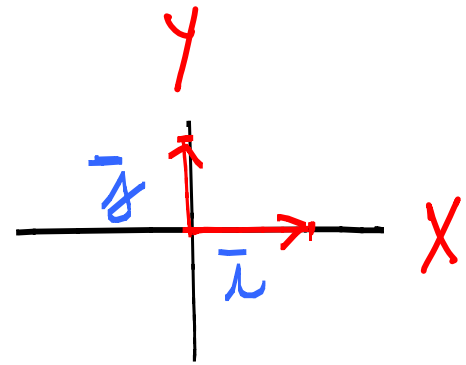
- Eenheidsvektor van $\bar{a}$: $\frac{1}{|\bar{a}|} \times \bar{a} = \frac{1}{5} \langle -3, 4 \rangle$

- Dan is $\vec{b} = \overset{\substack{\uparrow \\ \text{teencorgestelde richting}}}{-} (2) \left(\frac{1}{5} \right) \langle -3, 4 \rangle$
 $= -\frac{2}{5} \langle -3, 4 \rangle = \langle \frac{+6}{5}, \frac{-8}{5} \rangle$

Standard basis vektore:

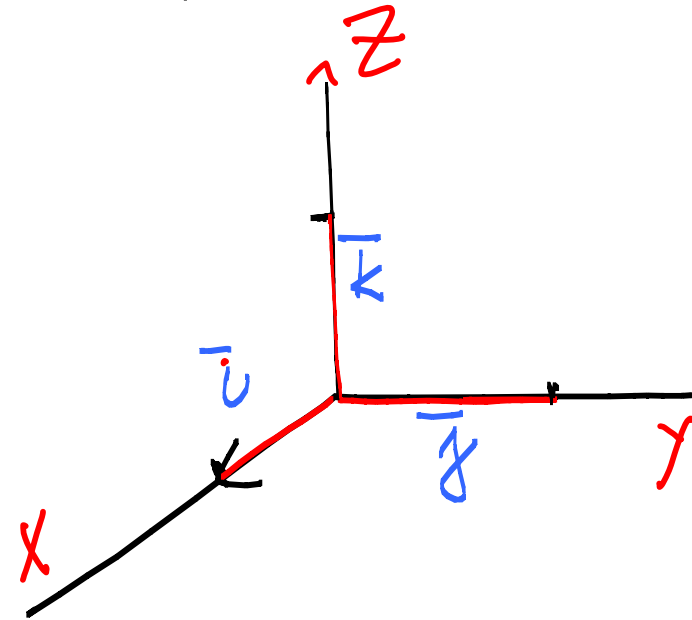
In $\mathbb{R}^2$:

- $\vec{i} = \langle 1, 0 \rangle$
- $\vec{j} = \langle 0, 1 \rangle$



In $\mathbb{R}^3$:

- $\vec{i} = \langle 1, 0, 0 \rangle$
- $\vec{j} = \langle 0, 1, 0 \rangle$
- $\vec{k} = \langle 0, 0, 1 \rangle$



* $\vec{j}_y$ kan enige vektor itu die
standard basis vektor skryf.

- $\langle a, b, c \rangle = \langle a, 0, 0 \rangle + \langle 0, b, 0 \rangle + \langle 0, 0, c \rangle$
 $= a \langle 1, 0, 0 \rangle + b \langle 0, 1, 0 \rangle + c \langle 0, 0, 1 \rangle$
 $= a\bar{i} + b\bar{j} + c\bar{k}$

- $\langle 3, -4 \rangle = \langle 2, 0 \rangle + \langle 0, -4 \rangle$
 $= 2 \langle 1, 0 \rangle - 4 \langle 0, 1 \rangle = 2\bar{i} - 4\bar{j}$

- $2\bar{i} + 3\bar{j} - 4\bar{k} = \langle 2, 3, -4 \rangle$

- $a\bar{a} = \bar{i} + 2\bar{j} - 3\bar{k}$
 $\bar{b} = 4\bar{i} + 7\bar{k}$

* Bepaal: $2\bar{a} + 3\bar{b} = 2(\bar{i} + 2\bar{j} - 3\bar{k}) + 3(4\bar{i} + 7\bar{k})$
 $= 2\bar{i} + 4\bar{j} - 6\bar{k} + 12\bar{i} + 21\bar{k} = \langle 14, 4, 15 \rangle$

$$\begin{aligned}
 \text{of } 2\bar{a} + 3\bar{b} &= 2\langle 1, 2, -3 \rangle + 3\langle 4, 0, 7 \rangle \\
 &= \langle 2, 4, -6 \rangle + \langle 12, 0, 21 \rangle \\
 &= \langle 14, 4, 15 \rangle
 \end{aligned}$$

• Bepaal 'n eenheidsvektor $\bar{b}$ in die rigting van die vektor $\bar{a} = 2\bar{i} - \bar{j} - 2\bar{k}$

$$\bullet |\bar{a}| = |\langle 2, -1, -2 \rangle| = \sqrt{9} = 3$$

• Eenheidsvektor

$$\text{is } \frac{\bar{a}}{|\bar{a}|} = \frac{1}{3} \langle 2, -1, -2 \rangle = \left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$$

$$\vec{b} = \left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$$

$$= \frac{2}{3} \vec{i} - \frac{1}{3} \vec{j} - \frac{2}{3} \vec{k}$$